
JHB Curtain Cleaning Specialists

Technical Handoff Document

Complete Website Build Documentation for AutoClaw

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Classification: Confidential — Business Critical

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1. Executive Summary

1.1 Project Overview

On The Spot Curtain Cleaning Specialists is Johannesburg's premier on-site curtain cleaning service, founded by Kathy Dunlop. This document covers the complete build of their production-ready website — a Next.js 16 application with 19 routes (1 homepage + 18 landing pages), AI-powered chatbot, multi-step booking wizard, POPIA-compliant cookie consent, and comprehensive SEO infrastructure including JSON-LD structured data.

1.2 Technology Stack

- **Framework:** Next.js 16 with App Router (TypeScript 5)
- **Styling:** Tailwind CSS 4 with shadcn/ui component library (New York style)
- **Animations:** Framer Motion 12
- **Icons:** Lucide React
- **Forms:** React Hook Form + Zod validation
- **AI:** z-ai-web-dev-sdk (LLM chatbot)
- **Database:** Prisma ORM with SQLite (available, configured)
- **State:** Zustand (available), TanStack Query (available)
- **Runtime:** Bun

1.3 Key Metrics

Metric	Value
Total Routes	19 (1 homepage + 6 service + 6 sector + 6 area)
Total Components	30+ React components
API Endpoints	4 (contact, newsletter, chat, health)
AI Features	Chatbot (active), Image Gen, TTS, Vision, Web Search (available)
SEO	JSON-LD structured data on every

Metric	Value
	page (LocalBusiness + Service + FAQ)
Performance	LCP ≤ 2.5s target, Mobile-First responsive
Compliance	POPIA (cookie consent, privacy policy, data handling)

2. Project Structure

2.1 Directory Architecture

```
src/
├── app/
│   ├── layout.tsx           # Root layout with fonts, JSON-LD, Toaster
│   ├── page.tsx            # Homepage (composes 16 sections)
│   └── globals.css         # Brand tokens, animations, utility
├── classes
│   ├── services/[slug]/
│   │   └── page.tsx        # Server component with
│   │                       generateStaticParams/generateMetadata
│   └── service-landing-client.tsx # Client component for service
├── pages
│   ├── sectors/[slug]/
│   │   ├── page.tsx        # Server component for sector pages
│   │   └── sector-landing-client.tsx # Client component
│   ├── areas/[slug]/
│   │   ├── page.tsx        # Server component for area pages
│   │   └── area-landing-client.tsx # Client component
│   └── api/
│       ├── route.ts        # Health check endpoint
│       ├── contact/route.ts # POST - Multi-step form submissions
│       ├── newsletter/route.ts # POST - Newsletter subscriptions
│       └── chat/route.ts    # POST - AI chatbot (z-ai-web-dev-sdk)
├── components/
│   ├── header.tsx          # Sticky header with dropdown navigation
│   ├── hero-section.tsx    # Homepage hero with metallic-sweep
│   ├── services-section.tsx # 6 service cards with landing page links
│   └── process-section.tsx  # 4-step process + guarantees + comparison
├── table
│   ├── sectors-section.tsx  # 6 sector cards with colored borders
│   ├── areas-section.tsx   # 6 area cards with suburb lists
│   ├── about-section.tsx   # Kathy Dunlop founder story
│   ├── faq-section.tsx     # 9-item accordion with JSON-LD FAQ schema
│   ├── newsletter-strip.tsx # Email capture for homepage
│   ├── contact-section.tsx # Multi-step quote wizard (4 steps)
│   └── cta-section.tsx     # CTA band
```

```
├── footer.tsx # 4-column footer with landing page links
├── whatsapp-button.tsx # Floating WhatsApp with pulse animation
├── cookie-consent.tsx # POPIA-compliant consent bar
├── ai-chatbot.tsx # AI chatbot widget with z-ai-web-dev-sdk
├── social-icons.tsx # Custom SVG social media icons
├── landing/
│   ├── landing-hero.tsx # Reusable hero for landing pages
│   ├── landing-sidebar.tsx # Reusable sidebar (assessment,
guarantees, contact)
│   ├── landing-cta.tsx # Reusable CTA band
│   └── landing-newsletter.tsx # Reusable newsletter strip
├── ui/ # 45+ shadcn/ui components
├── lib/
│   ├── site-data.ts # All website content as typed constants
(1,175 lines)
│   ├── utils.ts # Utility functions (cn helper)
│   └── db.ts # Prisma database client
├── hooks/
│   ├── use-toast.ts # Toast notification hook
│   └── use-mobile.ts # Mobile detection hook
├── prisma/
└── schema.prisma # Database schema
```

2.2 Public Assets

```
public/
├─ brand/
│   ├── logo.png           # Bronze seal logo (from Logo.png upload)
│   └─ bronze-seal.png     # Favicon
└─ robots.txt              # Search engine directives
```

3. Routing Architecture

3.1 Homepage (/)

Composes 16 components in sequence:

- 1. Header (fixed, glass-on-scroll, dropdown nav)
- 2. HeroSection (full-viewport, emerald gradient, metallic-sweep text)
- 3. ServicesSection (6 cards → link to /services/[slug])
- 4. ProcessSection (4-step timeline + 5 guarantees + comparison table)
- 5. SectorsSection (6 cards → link to /sectors/[slug])

- 6. AreasSection (6 cards → link to /areas/[slug])
- 7. AboutSection (Kathy Dunlop founder story + stats)
- 8. FAQSection (9-item accordion + JSON-LD FAQ schema)
- 9. NewsletterStrip (email capture)
- 10. ContactSection (multi-step form + contact sidebar)
- 11. CTASection (gradient CTA band)
- 12. Footer (4 columns: brand, sectors, services, contact)
- 13. WhatsAppButton (floating, pulse animation)
- 14. CookieConsent (POPIA-compliant)
- 15. AIChatbot (floating chat widget)

3.2 Service Landing Pages

(/services/[slug])

6 pages with SSG via generateStaticParams:

- /services/curtain-blind-cleaning (Primary — highlighted in nav)
- /services/mattress-sanitisation
- /services/upholstery-carpet-cleaning
- /services/master-guarding
- /services/fire-proofing
- /services/rug-care

Each page includes: Hero → Guarantee Badges → Main Content (included features, process steps, FAQ, data tables) + Sidebar (assessment, guarantees, contact) → Newsletter → CTA → Footer → WhatsApp → Cookie Consent

3.3 Sector Landing Pages

(/sectors/[slug])

6 pages with unique sector colors:

Route	Sector Color
/sectors/hotels	#0A369D Deep Blue

Route	Sector Color
/sectors/corporate	#082D84 Dark Blue
/sectors/healthcare	#059669 Green
/sectors/education	#D97706 Amber
/sectors/theatres	#6C3483 Purple
/sectors/residential	#0A369D Deep Blue

Each page includes: Hero (sector color) → Guarantee Badges → Key Stats → Decision-Maker Section (role-specific content) → Sector FAQ → Newsletter → CTA

3.4 Area Landing Pages (/areas/[slug])

6 pages with geo-targeted content:

Route	Focus
/areas/jhb-north	Highveld dust focus
/areas/jhb-east	O.R. Tambo hotel corridor
/areas/jhb-south	Heavy drapes specialists
/areas/jhb-west	Mining dust legacy
/areas/jhb-central	Heritage homes
/areas/pretoria-midrand	Jacaranda pollen

Each page includes: Hero → Focus Badge → Main Content (paragraphs + services table + suburb badges) + Sidebar → Newsletter → CTA

4. Brand Identity

4.1 Logo

Circular bronze medallion seal with:

- Outer ring: Bronze/brass gradient with stitched leather-patterned border
- Inner circle: Textured beige/parchment field
- Green draped curtains (center)
- Golden swoosh (cleaning motion arc)
- 5 teal stars
- Text: "JHB CURTAIN CLEANING" (top) / "ON-SITE CLEANING SPECIALISTS" (bottom)

4.2 Color System

Token	Value	Usage
Primary Emerald	#064e3b	Primary brand color, headers, CTAs
Primary Bronze	#a87d43	Accent color, highlights, nav active
Bronze Highlight	#c69c6d	Metallic text effects
Surface Base	#ffffff	Page background
Surface Ivory	#f9f9f6	Alternate section backgrounds
Surface Dim	#dadad7	Borders, dividers

Sector-Specific Colors

Sector	Color
Hotels & Hospitality	#0A369D
Corporate Offices	#082D84
Healthcare	#059669
Schools & Education	#D97706
Theatres & Venues	#6C3483
Residential	#0A369D

4.3 Typography

- **Body:** Inter (Google Fonts, display: swap)
- **Headings:** Playfair Display (Google Fonts, display: swap)
- **CSS variables:** --font-inter, --font-playfair

4.4 Animations

- **Metallic Sweep:** Gradient animation on "Johannesburg" hero text and 10% OFF badge
- **Trust Scroll:** Auto-scrolling trust strip
- **Service Card Hover:** translateY(-4px) + shadow
- **WhatsApp Pulse:** Pulse ring animation
- **Section Fade:** IntersectionObserver-based fade-in
- **Framer Motion:** Entrance animations, stagger effects, layout transitions

5. Data Architecture

5.1 Site Data (src/lib/site-data.ts)

All website content is stored as typed constants in a single 1,175-line file.
Key exports:

Export	Type	Description
SITE_CONFIG	Object	Company info, phones, email, address, social links, hours
TRUST_BADGES	Array	5 credibility badges
SERVICES	Array	6 services with descriptions, features, pricing
SECTORS	Array	6 sectors with colors, icons, features
AREAS	Array	6 areas with suburbs,

Export	Type	Description
		focus descriptions
PROCESS_STEPS	Array	4-step cleaning process
GUARANTEES	Array	5 customer guarantees
COMPARISON_TABLE	Object	10-row On-Site vs Conventional comparison
FAQS	Array	9 FAQs with answers
FOUNDER	Object	Kathy Dunlop bio, credentials, stats
NAV_LINKS	Array	7 navigation anchors
SERVICE_LANDING_DATA	Record	6 service landing pages (full content)
SECTOR_LANDING_DATA	Record	6 sector landing pages (full content)
AREA_LANDING_DATA	Record	6 area landing pages (full content)

5.2 Business Contact Details

Field	Value
Business Name	On The Spot Curtain Cleaning Specialists
WhatsApp/Mobile	075 011 9200
Office Phone	071 622 6723
Email	info@curtaincleaning.org
Address	Martha North Rd, Randburg, Johannesburg 2194, South Africa
Domain	curtaincleaning.org
Founder/MD	Kathy Dunlop
Hours	Mon-Fri 07:00-18:00, Sat 08:00-14:00

6. API Endpoints

6.1 POST /api/contact

Multi-step form submission handler.

- **Request body:** { sector, area, name, email, phone, propertyType, windowCount, notes, popiaConsent }
- **Validation:** Required fields check, email format, POPIA consent required
- **Response:** { success: true, message: "..." } or { error: "..." }

6.2 POST /api/newsletter

Newsletter subscription handler.

- **Request body:** { email }
- **Validation:** Email format check
- **Response:** { success: true, message: "..." } or { error: "..." }

6.3 POST /api/chat

AI chatbot endpoint using z-ai-web-dev-sdk.

- **Request body:** { message, history }
- **System prompt:** Curtain care assistant with full company knowledge
- **Response:** { message: "AI response" } or { error: "..." }

6.4 GET /api/route

Health check endpoint.

- **Response:** { status: "ok" }

7. AI Integrations

7.1 Active: AI Chatbot (z-ai-web-dev-sdk)

- Location: /api/chat (backend) + ai-chatbot.tsx (frontend widget)
- Model: z-ai-web-dev-sdk chat.completions.create()
- System prompt: Full company knowledge base (services, pricing, areas, guarantees)
- Features: Multi-turn conversation, typing indicator, markdown rendering
- Error fallback: Shows phone number for direct contact

7.2 Available (Not Yet Implemented)

These AI capabilities are available via z-ai-web-dev-sdk and can be activated:

Feature	SDK Method	Potential Use Case
Image Generation	zai.image.generations.create()	Generate before/after service images
Vision Analysis	zai.chat.completions.createVision()	Fabric photo analysis from customer uploads
Text-to-Speech	zai.audio.speech.create()	Voice FAQ answers, accessibility
Speech-to-Text	zai.audio.transcriptions.create()	Voice input for contact form
Web Search	(via CLI or SDK)	Live competitor pricing in chatbot

8. SEO Infrastructure

8.1 JSON-LD Structured Data

- Root layout: LocalBusiness schema with full address, geo coordinates, opening hours, offer catalog (6 services), aggregate rating
- Service pages: Service schema + FAQPage schema (per page)
- FAQ section: FAQPage schema with 9 Q&As
- All schemas use the Russian Doll pattern: LocalBusiness → hasOfferCatalog → Service

8.2 Metadata

- Homepage: Custom title, description, keywords (10), OpenGraph, Twitter Card, robots
- Landing pages: Dynamic generateMetadata() per slug with unique title/description/OpenGraph
- Favicon: /brand/bronze-seal.png

8.3 Performance Targets

Metric	Target	Status
LCP	$\leq 2.5s$	Configured
INP	$\leq 200ms$	Configured
CLS	≤ 0.1	Configured
Mobile-First	Required	Implemented

9. Compliance

9.1 POPIA (Protection of Personal Information Act)

- Cookie consent bar: "Accept All" / "Essential Only" options
- localStorage persistence of consent choice
- No tracking pixels before consent
- POPIA consent checkbox on contact form
- Privacy policy section referenced in cookie bar
- Full POPIA compliance documentation in original website copy

9.2 PAIA (Promotion of Access to Information Act)

- Referenced in footer: "POPIA Compliant"
- PAIA Section 51 manual referenced in technical specification
- Official PAIA Form 2 for information requisition

10. Deployment Guide

10.1 Recommended Platform: Vercel

- Zero-config for Next.js 16
- Global CDN (180+ edge locations)
- Automatic HTTPS/SSL
- Free tier sufficient for initial traffic
- Auto-deploys from GitHub

10.2 Deployment Steps

- 1. Push code to GitHub repository
- 2. Connect GitHub repo to Vercel
- 3. Configure environment variables (if any API keys needed)
- 4. Deploy (automatic)
- 5. Add custom domain: curtaincleaning.org
- 6. Configure DNS: A record @ → 76.76.21.21, CNAME www → cname.vercel-dns.com
- 7. SSL auto-provisioned by Vercel

10.3 DNS Configuration

Type	Name	Value
A	@	76.76.21.21
CNAME	www	cname.vercel-dns.com

10.4 Post-Launch Checklist

- 1. Submit sitemap to Google Search Console
- 2. Set up Google Business Profile
- 3. Configure professional email (info@curtaincleaning.org)
- 4. Set up Google Analytics 4
- 5. Test AI chatbot in production
- 6. Run Lighthouse audit
- 7. Configure Vercel Speed Insights for CWV monitoring

10.5 Estimated Costs

Item	Cost
Domain (curtaincleaning.org)	R150-R250/year
Vercel Hosting	Free (Pro: \$20/mo if needed)
SSL Certificate	Free (Vercel)

Item	Cost
Professional Email	R50-R150/month
AI Chatbot (production)	R300-R500/month
Total Annual	R5,000-R10,000

11. Social Media Accounts

The following social media accounts are configured in the site data. These should be registered/claimed if not already:

Platform	URL	Status
Facebook	facebook.com/ curtaincleaning	Needs verification
Instagram	instagram.com/ curtaincleaning	Needs verification
YouTube	youtube.com/ @curtaincleaning	Needs verification
LinkedIn	linkedin.com/company/ curtaincleaning	Needs verification
TikTok	tiktok.com/ @curtaincleaning	Needs verification
Pinterest	pinterest.com/ curtaincleaning	Needs verification
X/Twitter	twitter.com/ curtaincleaning	Needs verification

12. Third-Party Configurations

12.1 Stitch (Google)

- **Stitch API Key:** AQ.Ab8RN6I52VtWs7F2O-h1Qtrr-0nZrMSojuGF6kOpq3INT0ihkA
- **Stitch Project URL:**
<https://stitch.withgoogle.com/projects/13681111782633686811>
- **Purpose:** Data integration (referenced in technical specification)

12.2 Google Services Needed

- Google Search Console (for sitemap submission)
- Google Business Profile (for local SEO / map pack)
- Google Analytics 4 (for traffic analysis)
- Google Workspace (for professional email)

13. Known Considerations & Future Enhancements

13.1 Immediate Actions Needed

- 1. Register domain curtaincleaning.org
- 2. Set up Vercel account and deploy
- 3. Configure professional email
- 4. Register/verify all social media accounts
- 5. Set up Google Business Profile
- 6. Test AI chatbot with production API key

13.2 Recommended Enhancements

- 1. Smart Fabric Analyzer: Customers upload curtain photos → AI identifies fabric type
- 2. AI Image Generation: Professional before/after service images
- 3. Voice FAQ: Text-to-speech on FAQ answers for accessibility
- 4. Live Price Chatbot: Enhanced chatbot with web search for competitor pricing
- 5. Google Maps integration on area pages
- 6. Testimonials/reviews carousel from Google Reviews
- 7. Blog section for content marketing and SEO
- 8. Exit-intent popup for email capture

13.3 Technical Debt

- Database (Prisma/SQLite) is configured but not actively used — contact form currently logs to console
- Sitemap generation (sitemap.ts) referenced in specs but not yet implemented
- robots.txt exists but should be enhanced for production
- Environment variables needed for production API keys

Appendix A: File Manifest

File Path	Purpose
src/app/layout.tsx	Root layout with fonts, JSON-LD, Toaster
src/app/page.tsx	Homepage — composes 16 sections
src/app/globals.css	Brand tokens, animations, utility classes
src/app/services/[slug]/page.tsx	Service page server component
src/app/services/[slug]/service-landing-client.tsx	Service page client component
src/app/sectors/[slug]/page.tsx	Sector page server component

File Path	Purpose
src/app/sectors/[slug]/sector-landing-client.tsx	Sector page client component
src/app/areas/[slug]/page.tsx	Area page server component
src/app/areas/[slug]/area-landing-client.tsx	Area page client component
src/app/api/route.ts	Health check endpoint
src/app/api/contact/route.ts	Contact form POST handler
src/app/api/newsletter/route.ts	Newsletter subscription POST handler
src/app/api/chat/route.ts	AI chatbot POST handler
src/components/header.tsx	Sticky header with dropdown navigation
src/components/hero-section.tsx	Homepage hero with metallic-sweep
src/components/services-section.tsx	6 service cards with landing page links
src/components/process-section.tsx	4-step process + guarantees + comparison table
src/components/sectors-section.tsx	6 sector cards with colored borders
src/components/areas-section.tsx	6 area cards with suburb lists
src/components/about-section.tsx	Kathy Dunlop founder story
src/components/faq-section.tsx	9-item accordion with JSON-LD FAQ schema
src/components/newsletter-strip.tsx	Email capture for homepage
src/components/contact-section.tsx	Multi-step quote wizard (4 steps)
src/components/cta-section.tsx	CTA band
src/components/footer.tsx	4-column footer with landing page links
src/components/whatsapp-button.tsx	Floating WhatsApp with pulse animation
src/components/cookie-consent.tsx	POPIA-compliant consent bar
src/components/ai-chatbot.tsx	AI chatbot widget with z-ai-web-dev-sdk
src/components/social-icons.tsx	Custom SVG social media icons

File Path	Purpose
src/components/landing/landing-hero.tsx	Reusable hero for landing pages
src/components/landing/landing-sidebar.tsx	Reusable sidebar (assessment, guarantees, contact)
src/components/landing/landing-cta.tsx	Reusable CTA band
src/components/landing/landing-newsletter.tsx	Reusable newsletter strip
src/components/ui/*	45+ shadcn/ui components
src/lib/site-data.ts	All website content as typed constants (1,175 lines)
src/lib/utils.ts	Utility functions (cn helper)
src/lib/db.ts	Prisma database client
src/hooks/use-toast.ts	Toast notification hook
src/hooks/use-mobile.ts	Mobile detection hook
prisma/schema.prisma	Database schema
public/brand/logo.png	Bronze seal logo
public/brand/bronze-seal.png	Favicon
public/robots.txt	Search engine directives

Appendix B: Component Dependency Map

The following shows which components import from where:

Homepage (src/app/page.tsx)

Imports: Header, HeroSection, ServicesSection, ProcessSection, SectorsSection, AreasSection, AboutSection, FAQSection, NewsletterStrip, ContactSection, CTASection, Footer, WhatsAppButton, CookieConsent, AIChatbot

Service Pages (service-landing-client.tsx)

```
Imports: Header, LandingHero, LandingSidebar, LandingCTA,  
        LandingNewsletter, Footer, WhatsAppButton, CookieConsent,  
        AIChatbot
```

Sector Pages (sector-landing-client.tsx)

```
Imports: Header, LandingHero, LandingSidebar, LandingCTA,  
        LandingNewsletter, Footer, WhatsAppButton, CookieConsent,  
        AIChatbot
```

Area Pages (area-landing-client.tsx)

```
Imports: Header, LandingHero, LandingSidebar, LandingCTA,  
        LandingNewsletter, Footer, WhatsAppButton, CookieConsent,  
        AIChatbot
```

Header Component

```
Imports: Link (next/link), MobileMenu (inline), NAV_LINKS (site-data)  
Depends on: site-data.ts for navigation links
```

Landing Components

```
landing-hero.tsx → imports: Link, SERVICES/SECTORS/AREAS (site-data)  
landing-sidebar.tsx → imports: GUARANTEES, SITE_CONFIG (site-data)  
landing-cta.tsx → imports: Link, SITE_CONFIG (site-data)  
landing-newsletter.tsx → imports: useState (React)
```

Data Flow

```
site-data.ts → All page components (content source of truth)  
site-data.ts → Server components (generateMetadata,  
generateStaticParams)  
site-data.ts → Client components (display data)  
API routes → z-ai-web-dev-sdk (chat endpoint)  
API routes → Console logging (contact, newsletter)
```

Appendix C: Data Schema

Full type definitions from site-data.ts for landing page data structures:

SERVICE_LANDING_DATA Structure

```
type ServiceLandingData = {
  slug: string;
  title: string;
  heroSubtitle: string;
  description: string;
  metaTitle: string;
  metaDescription: string;
  includedFeatures: string[];
  processSteps: { step: number; title: string; description: string }[];
  faqs: { question: string; answer: string }[];
  dataTables: { title: string; headers: string[]; rows: string[][] }[];
  ctaText: string;
};
```

SECTOR_LANDING_DATA Structure

```
type SectorLandingData = {
  slug: string;
  title: string;
  heroSubtitle: string;
  sectorColor: string;
  metaTitle: string;
  metaDescription: string;
  keyStats: { label: string; value: string }[];
  decisionMakers: {
    role: string;
    concerns: string[];
    solution: string;
  }[];
  sectorFaqs: { question: string; answer: string }[];
  ctaText: string;
};
```

AREA_LANDING_DATA Structure

```
type AreaLandingData = {
  slug: string;
  title: string;
  heroSubtitle: string;
  focusLabel: string;
  metaTitle: string;
```

```
metaDescription: string;
mainContent: string[];
servicesTable: { service: string; description: string }[];
suburbs: string[];
ctaText: string;
};
```

SITE_CONFIG Structure

```
type SiteConfig = {
  name: string;
  tagline: string;
  phone: string;
  whatsapp: string;
  email: string;
  address: string;
  domain: string;
  social: {
    facebook: string;
    instagram: string;
    youtube: string;
    linkedin: string;
    tiktok: string;
    pinterest: string;
    twitter: string;
  };
  hours: { days: string; time: string }[];
  founder: { name: string; title: string };
};
```

— *End of Document* —

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